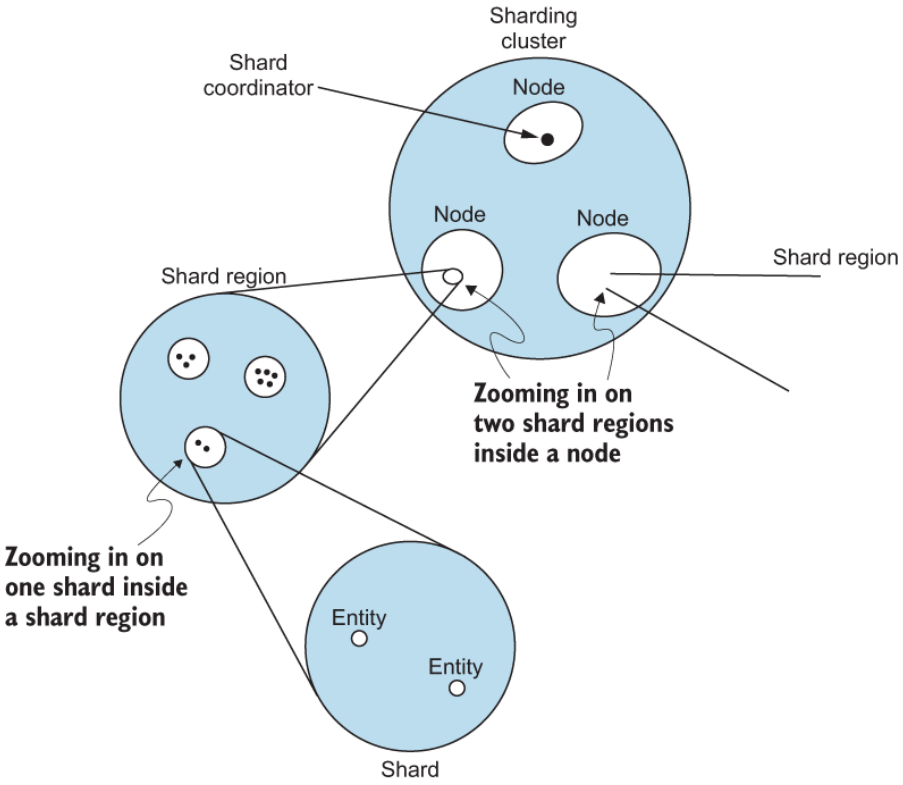


Figure 9.2 Components of Akka cluster sharding

There is only one SC in the entire cluster, and it has two main tasks: monitoring the state of the shards and facilitating access to each entity. SR monitoring is necessary to ensure that shards are moved from a congested or unhealthy region to a healthy one. This means an unreachable node must be removed from the cluster before the shard and its entities can be moved. Otherwise, a previously unreachable node could rejoin the cluster after rebalancing, and there would be duplicates in the cluster. Fortunately, the Split Brain Resolver (SBR) manages unreachable nodes (more on the SBR in chapter 13).

The SC provides the necessary mechanism to communicate with entities. The same function used to assign entities is also used to find them; however, this is done indirectly. When a message is sent to an entity, it must be done through an SR. If this SR does not know where the entity lives, and in which node, it asks the SC. You might think it could lead to a bottleneck if the SC received a query for every message, but this is not the case. Once the SC finds the correct SR to forward the message, it passes this information to all SRs in the cluster and delegates all further forwarding to the entities to the SRs.

Through the API, you can send messages to an entity two ways. The first technique is indirect: send the message to the SR of the same node. If the entity is in a different node—that is, in a different SR—the local SR forwards the message to the SR where the entity is located, and the latter forwards the message to the entity itself. The second technique is to send messages